

2009 HSC Notes from the Marking Centre – Chemistry

Contents

- [Section I – Core](#)
- [Section II – Options](#)

Introduction

This document has been produced for the teachers and candidates of the Stage 6 course in Chemistry. It contains comments on candidate responses to the 2009 Higher School Certificate examination, indicating the quality of the responses and highlighting their relative strengths and weaknesses.

This document should be read along with the relevant syllabus, the 2009 Higher School Certificate examination, the marking guidelines and other support documents which have been developed by the Board of Studies to assist in the teaching and learning of Chemistry.

Teachers and students are advised that, in December 2008, the Board of Studies approved changes to the examination specifications and assessment requirements for a number of courses. These changes will be implemented for the 2010 HSC cohort. [Information on a course-by-course basis](#) is available on the Board's website.

General Comments

Teachers and candidates should be aware that examiners may ask questions that address the syllabus outcomes in a manner that requires candidates to respond by integrating their knowledge, understanding and skills developed through studying the course including the prescribed focus areas. It is important to understand that the Preliminary course is assumed knowledge for the HSC course.

Candidates need to be aware that the mark allocated to the question and the answer space (where this is provided on the examination paper) are a guide to the length of the required response. A longer response will not in itself lead to higher marks. Writing far beyond the indicated space may reduce the time available for answering other questions.

Candidates need to be familiar with the Board's [Glossary of Key Words](#) which contains some terms commonly used in examination questions. However, candidates should also be aware that not all questions will start with or contain one of the key words from the glossary. Questions such as 'how?', 'why?' or 'to what extent?' may be asked or verbs may be used which are not included in the glossary, such as 'design', 'translate' or 'list'.

Teachers and candidates are reminded that mandatory skills content in Module 9.1 is examinable in both the Core and Option questions.

Candidates should use examination time to analyse the question, plan their responses carefully, and then work within that framework to produce clear, logical and concise responses. The response may include the use of dot points, diagrams and/or tables and planning the response will help to avoid internal contradictions. Holistic responses need to be logical, well constructed and relevant to the questions asked.

In better responses, candidates:

- set out all working for numerical questions
- thought carefully about the units to be used and the quantities to be substituted into formulae
- did not repeat the question as part of the response
- looked at the structure of the whole question and noted that in some questions the parts followed on from each other, eg responses in part (a) led to the required response in part (b)
- used appropriate equipment, eg pencils and a ruler to draw graphs. (A clear plastic ruler would aid candidates to plot points that are further from the axes and rule straight lines of best fit where relevant.)
- accurately transcribed and used values from the periodic table and data sheet for calculations
- included balanced chemical equations where appropriate.

In Section II, each Option question is divided into a number of parts. Candidates should clearly label each part of the question when writing in their answer booklets. In part (e) of the 2009 Option questions, the best responses presented ideas coherently and logically and included the correct use of scientific principles and ideas. Candidates are strongly advised to answer the Option they have studied in class.

Section I – Core

Part B

Question 16

Better responses included the quantity and name of the reagents including concentrated H_2SO_4 . Better responses also clearly provided a relevant safety precaution. Weaker responses drew the condenser but did not do so correctly.

Question 17

Mid-range responses stated that water is a polar molecule and ethanol is a polar and a non-polar molecule. Better responses related these properties to the molecular structure and stated that dispersion forces are in action at the non-polar end of ethanol. While weaker responses stated that hydrogen bonding occurred between the solvent and the solute, they did not clearly show the hydrogen bonding between the molecules. Some weaker responses showed confusion between intramolecular and intermolecular bonding.

Question 18

The better responses stated that oxides of nitrogen (NO_x) are formed by the reaction between nitrogen gas (N_2) and oxygen gas at high temperatures associated with combustion in coal-fired power stations and internal-combustion engines such as cars.

These responses included an assessment of direct evidence of the increase in NO_x , eg gas bubbles in ice-cores in Antarctica, and indirect evidence such as increase in brown haze due to photochemical smog. The better responses discussed at least two effects of NO_x (photochemical smog, respiratory problems, formation of acid rain and effects of acid rain) in the atmosphere and justified the need to monitor.

Better responses included at least two relevant balanced chemical equations. The better responses included the equation for the reaction between nitrogen dioxide and water forming nitric acid and nitrous acid.

Weaker responses did not identify the formation of photochemical smog as an effect of NO_x and only identified effects of acid rain. Weaker responses did not provide evidence of increase of NO_x , reasons for monitoring, an assessment or chemical equations. There was a common misconception that nitrogen, like sulfur, is a contaminant in coal and petrol. Some weaker responses appeared to be prepared answers; they included information about oxides that were inappropriate as a response to this question.

Question 19

Better responses included the raw materials and outlined the physical and chemical processes for the production of the polymer product from the raw materials with equations.

Better responses included correct balanced chemical equations for the polymerisation process and the formation of the monomer selected for the polymer, depending on the pathway selected from the raw material.

Weaker responses did not include distillation as a physical process of separation and outlined the polymerisation process of the selected monomers in words, rather than by using a chemical equation.

Question 20

- a. The better responses set out the relevant working and distinguished between the mass of water and the mass of ethanol. These responses calculated q and determined the number of moles using $n = q/\Delta H$. Weaker responses confused the two mass values, did not correctly calculate the molar mass of ethanol and were confused about energy units.

Weaker responses confused relationships such as $\Delta H = q/n$ with $n = \Delta H/q$.

- b. The best responses clearly stated two methods of reducing heat loss. Weaker responses were uncertain about conductivity and insulation. Some responses included heat of solution or heat of neutralisation and suggested using a polystyrene cup as a calorimeter.

Question 21

- a. Better responses showed evidence of planning and correct use of chemistry vocabulary. These responses clearly separated the answer to concentration from the answer for strength. These responses clearly stated features from the graph to support their answers.
- b. Better responses used the name of a salt of a weak monoprotic acid as called for in the question, rather than the formula. Weaker responses included the salts of strong acids or of weak acids that are unsuitable for titration.
- c. Better responses calculated the concentration of H_3O^+ ions in solution and included correct standard units. Weaker responses did not accurately calculate the concentration of H_3O^+ ions in solution.
- d. Better responses included the correct range of the indicator and referred to the equivalence point on the graphs for both acids. Weaker responses were overly general or did not include the correct range of the indicator or did not accurately refer to the graph.

Question 22

- a. Better responses constructed a balanced equation for the neutralisation of hydrochloric acid with sodium hydroxide and included a balanced equation for the reaction between ammonia and acid. Better responses included correct states of matter. Weaker responses showed a lack of familiarity with the correct formula for ammonia.
- b. & c. Weaker responses did not follow the sequence of reactions and did not demonstrate an understanding that the excess acid from the first reaction was used in the second calculation. Better responses showed all working in questions requiring calculations as well as the correct answer. Weaker responses incorrectly extracted data from the stimulus.
- d. Better responses did not round off until the percentage of nitrogen in bread was calculated. These responses reported the final percentage correct to three significant figures, while weaker responses did not show the correct number of significant figures. Weaker responses used incorrect molar mass to calculate the mass of nitrogen; some included the molar mass of ammonia.

Question 23

Better responses demonstrated an understanding of the relationship between the shape of the graph and the reasons behind the shifts in terms of equilibrium and Le Chatelier's Principle. The best responses linked the physical changes (at all three times) to these reasons and identified the times at which the system was at equilibrium.

Weaker responses could not fully explain the reasons for the shift in equilibrium, with many simply stating 'according to Le Chatelier's Principle' while others omitted a reason altogether. Of the weaker responses that detailed the reasons for the shifts, some did not identify the physical changes responsible for the shifts. Some weaker responses tried to identify several possible changes at each time and ended up contradicting themselves and others failed to identify times at which these changes occurred. Some weaker responses tried to explain each component individually instead of treating the reaction as an equilibrium system. Other weaker responses showed evidence of rote learning rather than an understanding of graphs in answering the question asked.

Question 24

Better responses addressed all three aspects of the question. Clear indication of how metals absorb light was discussed, this included reference to the specific lamp used, how electrons are excited and referred to the amount of light absorbed. Better responses connected amount of absorbance with a concentration or calibration curve. Fully labelled diagrams were included. The use of AAS to monitoring specific environments for a named metal was included with reference to its application and the need for sensitivity.

Mid-range responses tried to address all aspects of the question but only did so superficially. Weaker responses provided incomplete or unlabelled diagrams and were generalised AAS and AES or flame with some showing confusion between tests.

Question 25

- a. Better responses correctly determined the average absorbance of phosphate from the three values with the appropriate number of three significant places and correctly used the graph to determine the concentration of phosphate in all three streams within an acceptable range.

Mid-range responses correctly determined the average absorbance of all three streams but did not accurately determine the correct phosphate concentration for all three streams. Mid-range responses did not make accurate readings of the graph for one or two of the streams, but not all three streams.

Weaker responses did not use all three values in determining the average phosphate absorbance or selected a value from the range of numbers provided in the question.

- b. Better responses compared the concentration of all three streams to the recommended maximum level for phosphate or compared two of the streams to the recommended maximum level and made a relative comparison for the third stream to the other two. Better responses provided at least two valid sources of phosphate and used terms such as 'runoff' to show how the sources can enter the streams.

Mid-range responses compared the phosphate concentration of all three streams to the valid maximum level and provided at least two valid sources of phosphate but did not show how these can elevate levels of phosphate in a stream.

Weaker responses either only compared the three streams to the maximum recommended level or they only provided sources of phosphate. Weak responses compared the three streams but made no references to the maximum accepted level of phosphate.

- c. Better responses included eutrophication as a water quality issue and stated that high levels of phosphate can lead to eutrophication of bodies of water. Better responses also described effects of eutrophication or could describe the eutrophication process.

Mid-range responses stated that either high levels of phosphate led to eutrophication or described an effect of eutrophication. Weaker responses provided effects to water systems which are non-specific to higher phosphate concentrations.

Question 26

- a. In better responses, candidates read the instructions carefully and wrote the reduction half equation for Fe^{3+} to Fe^{2+} and a reduction half equation for Pb^{2+} to Pb. The better responses gave the resulting net ionic equation after adding the two half equations. Weaker responses wrote the half equation for Fe to Fe^{2+} or Fe^{3+} or for platinum.
- b. Weaker responses multiplied $E^\ominus$ values or did not reverse the sign of $E^\ominus$ when reversing the half equation.
- c. Better responses labelled the electrodes as the anode and the cathode while weaker responses labelled the side of the beaker or the side of the cell. Weaker responses did not identify that oxidation occurred at the anode and reduction at the cathode.
- d. Better responses identified the formula or name of a suitable chemical to use as the salt bridge, with weaker responses giving inappropriate chemicals due to solubility issues with lead ions.

Section II – Options

Question 27 – Industrial Chemistry

- a. i. The better responses identified that the major use of sulfuric acid is in the production of superphosphate fertiliser.
- ii. The better responses sketched all three stages in the conversion of sulfur into sulfuric acid and included balanced chemical equations and appropriate conditions for the conversion of sulfur dioxide into sulfur trioxide. A common misconception was that oleum is aqueous.
- iii. The better responses identified two properties of sulfuric acid and directly related each to a hazard and an appropriate safety

precaution for transporting sulfuric acid. Weaker responses used vague terms such as dangerous rather than identifying specific hazardous properties.

- b. ii. The better responses showed clearly how the equilibrium concentrations of the reactant and products were determined and correctly substituted into the expression for K. Weaker responses did not show adequate working.
- iii. The better responses also stated that the value of K increases.
- c. The better responses identified a specific emulsion (eg mayonnaise) and related its use to its properties as an emulsion.
- d. i. Better responses used a labelled diagram to assist their explanation and used correct chemical terminology. Weaker responses gave a general description of the process only.
- e. The better responses demonstrated knowledge of the relevant chemical equations for each stage of the flowchart and displayed a good understanding of the Solvay process and the significant factors contributing to the location of the industrial plant. These responses were cohesive and well structured.

Question 28 – Shipwrecks, Corrosion and Conservation

- a. ii. The better responses described protection methods such as Sn plating, polymer coatings and Al_2O_3 coatings; which were relevant to food/general household chemical storage and compared the coatings.
- b. The better responses explained the protection provided in terms of the passivating surface formed on the zinc and its cathodic protection of iron.
- c. ii. The better responses used the E^0 values or relevant equations to rank the metals and stated the consequences of galvanic coupling for both metal combinations.
- d. i. Weaker responses identified the cathode/negative terminal. Weaker responses referred to the iron being reduced rather than using terms like cathode, anode or negative or positive terminal.
- ii. Better responses used a silver anode and solution of silver ions to electroplate the nickel spoon. Weaker responses treated the plating of silver on to the nickel as a metal displacement reaction.
- e. Better responses described the work of scientists and explained how their work led to a better understanding of electron transfer reactions. These responses were cohesive and well structured. Weaker responses described the work of Galvani and Volta but failed to demonstrate how their work led to a better understanding of electron transfer reactions.

Question 29 – Biochemistry of Movement

- a. i. Better responses identified a correct factor and also indicated that a change in that factor was needed. Weaker responses provided a factor, for example temperature, but did not qualify that it was a change in temperature that was required.
- ii. Better responses identified that hydrogen bonds, electrostatic/disulphide bridges and/or hydrophobic forces were responsible for the shape and thus the structure at the secondary and tertiary levels. Weaker responses incorrectly identified the peptide bond in the primary structure as being the first part of the protein that denatures.
- b. Better responses used all the data from the table to explain that polyphenoloxidase (PPO) only works at a specific, narrow pH range. Weaker responses did not mention any of the colour change or failed to account for the light red and medium red colours at pH lower than 6.
- c. i. Weaker responses did not include the energy term. Better responses recognised the graph as an energy profile diagram and calculate the ΔH .
- ii. Some weaker responses did not make the link between moles and grams and did not calculate a correct answer.
- iii. Weaker responses did not manipulate the simple numeric equation to calculate the correct answer.
- d. i. Better responses identified that the polarity of the molecule was due to the presence of the three hydroxyl groups. They established the role that hydroxyl groups play in hydrogen bonding, both with water (solubility) and with other glycerol molecules (viscosity). Weaker responses confused glycerol with glucose or confused hydrogen bonds with dispersion forces.
- ii. Stronger responses identified the dominance of the long non-polar hydrocarbon tails as being the factor leading to the insoluble nature of both fatty acids and triacylglycerols. Better responses linked this to the fact that the loss of hydroxyl groups during esterification meant that triacylglycerols lost much of their polarity. Weaker responses confused the terms polar/non-polar and hydrophilic/hydrophobic and as a result their explanations were contradictory.
- e. Some better responses gave a holistic and detailed explanation of the metabolic processes mentioned in the flow chart. These better responses included a selection of appropriate and correct equations, as asked for in the question. Stronger responses then linked this information when answering the section on training programs. These responses were cohesive and well structured. Weaker responses gave details of some appropriate metabolic process but failed to provide any appropriate equations or to link their response to training

programs.

Question 30 – Chemistry of Art

- a. iii. Weaker responses generally explained the basic features of the bonding between EDTA and Cu^{2+} including the formation of coordinate covalent bonds and donation of electron pairs by the ligand. Better responses included knowledge of how donated electron pairs occupy empty orbitals in the Cu^{2+} .
- b. i. Weaker responses confused the term orbital and subshell, and incorrectly stated the maximum electrons held in a p orbital as being 6.
- c. Weaker responses demonstrated a basic knowledge of ionisation energy and explained at least one feature of the graph but attempted to explain the trends in terms of stability of electron configurations, without considering factors such as increasing nuclear charge, new shell formation and shielding effects of inner electrons. Better responses clearly explained the general increase in first ionisation energy across a period, the reason for significant drops from elements 2-3, 10-11 and 18-19, and the reason for a decrease as you move down a group. Many also explained some minor irregularities in data within a period using extensive knowledge of subshell formation and Hund's Rule.
- d. i. Better responses explained spectra by incorporating the quantum concept to the atom and clearly relating spectral line formation to quantum drops between levels after initial excitation. Better responses successfully used diagrams successfully to support the answer.
ii. Better responses described how they would test the water sample using a flame test, starting with evaporation of the solution. They also emphasised the need to heat the sample in a non-luminous or blue flame. Responses included a large number of flame test methods.
- e. The best responses included specific references to pigments containing first series transition metals, including chemical formula and name of the mineral sources of these pigments. The better responses also described several uses of pigments by ancient civilisations. Weaker responses often showed evidence of insufficient planning, resulting in a lack of coherence and the addition of irrelevant material. The better responses related knowledge of toxicity of metal-based pigments or refinement/synthesis techniques to the development of newer synthetic pigments. These responses were cohesive and well structured.

Question 31 – Forensic Chemistry

- a. ii. Better responses showed a clear distinction between structures with increasing level of complexity.
iii. Better responses described the principle behind electrophoresis, with a description of the process and a comparison statement. Weaker responses discussed the comparison of food with no reference to proteins or the process of separation.
- b. ii. Better responses identified what the peaks represented, rather than how they got there. Weaker responses stated that the peaks represented the concentration of the component rather than stating that the height of the peak or intensity was proportional to the concentration of the component.
- c. Better responses considered the features of chromatography that related to small samples, rather than those that described the process and the specific features of the instrument.
- d. i. Weaker responses gave the principles of paper chromatography rather than the method.
ii. Weaker responses showed a misconception of the decolourisation of the alkene rather than the bromine water.
- e. Weaker responses linked incorrect samples with the instruments used to test them or used simple laboratory tests and did not include instrumental analysis. Better responses followed a logical argument and were succinct in their discussion.

2010023